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Explanation of On Variational Bounds of Mutual Information

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Explanation of the paper *On Variational Bounds of Mutual Information*, ICML 2019.

1. Introduction

This paper introduces how to derive class of variational bounds of the mutual information, including [MINE-f](#) and [InfoNCE](#). This paper also propose a new upper and a new lower bounds trading off between the bias and variance of the estimation of the MI.

Let's begin with this paper, [On Variational Bounds of Mutual Information](#), in ICML 2019.

2. Review of Variational Bounds of MI

MI:

$$I(X; Y) = \mathbb{E}_{p(x,y)} \left[\log \frac{p(x | y)}{p(x)} \right] = \mathbb{E}_{p(x,y)} \left[\log \frac{p(y | x)}{p(y)} \right] \quad (1)$$

Notes: In the following, there is a term “(un)normalized”, which indicates whether a variational distribution is normalized.

2.1. Normalized Upper and Lower Bounds

In Eq. (1), we can easily obtain different types of bounds with KL-divergence when we are substituting different variational distributions to different marginal or conditional distributions.

For example, if $p(y | x)$ is like an encoder in VAE, by inserting $q(y)$, which approximates $p(y)$, we can have a tractable variational upper bound:

$$\begin{aligned} I(X; Y) &\equiv \mathbb{E}_{p(x,y)} \left[\log \frac{p(y | x)}{p(y)} \right] \\ &= \mathbb{E}_{p(x,y)} \left[\log \frac{p(y | x)q(y)}{q(y)p(y)} \right] \\ &= \mathbb{E}_{p(x,y)} \left[\log \frac{p(y | x)}{q(y)} \right] - KL(p(y) || q(y)) \\ &\leq \mathbb{E}_{p(x)} [KL(p(y | x) || q(y))], \end{aligned}$$

where the inequality comes from the non-negative KL term. This bound is tight when $q(y) = p(y)$. VAE just uses the same thing in the optimization.

2.1.1. I_{BA}

We can do the opposite direction by using another variational distribution $q(x | y)$. The lower bound is derived by Barber & Agakov in 2003:

$$\begin{aligned} I(X; Y) &= \mathbb{E}_{p(x,y)} \left[\log \frac{q(x | y)}{p(x)} \right] \\ &\quad + \mathbb{E}_{p(y)} [KL(p(x | y) || q(x | y))] \\ &\geq \mathbb{E}_{p(x,y)} [\log q(x | y)] + h(X) \triangleq I_{BA}, \end{aligned}$$

where $q(x | y)$ can be thought of as a decoder and $h(X) = -\mathbb{E}_{p(x,y)} \log p(x)$ is the differential entropy of X . This bound is tight when $q(x | y) = p(x | y)$. The differential entropy term actually makes this bound intractable because $p(x)$ is always unknown. A further drawback of this bound is that the decoder term, is always challenging when the data is high-dimensional.

2.2. Unnormalized Lower Bounds

In this section, the lower bound is mainly studied with an unnormalized variational distribution $q(x | y)$.

The variational family is an energy-based style with a *critic* function $f(x, y)$ and scaled by the data density $p(x)$:

$$q(x | y) = \frac{p(x)}{Z(y)} e^{f(x,y)}, \text{ where } Z(y) = \mathbb{E}_{p(x)} [e^{f(x,y)}], \quad (2)$$

which is exactly what we did in [MINE](#).

2.2.1. I_{UBA}

Substituting $q(x | y)$ in I_{BA} , we can get an unnormalized lower bound I_{UBA} :

$$\begin{aligned} I_{UBA} &\triangleq \mathbb{E}_{p(x,y)} [\log q(x | y)] + h(X) \\ &= \mathbb{E}_{p(x,y)} [\log p(x) - \log Z(y) + f(x, y)] + h(X) \\ &= \mathbb{E}_{p(x,y)} [f(x, y)] - \mathbb{E}_{p(y)} [\log Z(y)]. \end{aligned}$$

This bound is tight when:

$$\begin{aligned}
q(x | y) &= \frac{p(x)}{Z(y)} e^{f(x,y)} = p(x | y) = \frac{p(x, y)}{p(y)} \\
f(x, y) &= \log \frac{p(x, y)}{p(x)} + \log \frac{Z(y)}{p(y)} \\
&= \log p(y | x) + c(y),
\end{aligned}$$

where $c(y) = \log \frac{Z(y)}{p(y)}$ is a function solely related y . This bound removes the intractable term $h(X)$ in I_{BA} ! Are we done? Not yet because the log partition function $\log Z(y) = \log \mathbb{E}_{p(x)} [e^{f(x,y)}]$ is still intractable. Can't we just use Monte Carlo to estimate it? "Yes, we can!" But such an estimation is biased. In a batch of data samples, we are actually calculating the mean of $\log e^{f(x,y)}$ because with each discrete data point, we can only calculate $\log \mathbb{E}_{p(x)} [e^{f(x,y)}]$ without the expectation. And until we calculate without the expectation term over the batch, we can finally calculate the mean of this term, which becomes $\mathbb{E}_{p(x)} [\log e^{f(x,y)}]$. Therefore, it is biased and intractable with Monte Carlo estimation.

2.2.2. I_{DV} and MINE

Applying Jensen's inequality to I_{UBA} can recover the bound of Donsker & Varadhan in 1983:

$$I_{UBA} \geq \mathbb{E}_{p(x,y)} [f(x, y)] - \log \mathbb{E}_{p(y)} [Z(y)] \triangleq I_{DV}. \quad (3)$$

It is still intractable because of the same biased reason.

If we apply Jensen's inequality in the other direction, we can have a tractable objective, but not a bound:

$$I \geq I_{UBA} \leq \mathbb{E}_{p(x,y)} [f(x, y)] - \mathbb{E}_{p(x)} [f(x, y)], \quad (4)$$

which is the same case when estimating I_{DV} using Monte Carlo. [MINE](#) is doing the same thing, which is not a bound at all!

2.2.3. I_{TUBA}

So, how can we obtain a tractable unnormalized lower bound? We need to escape from expectation inside log. Note that $\log(x) \leq \frac{x}{a} + \log(a) - 1, \forall x, a > 0$. Applying it to $\log Z(y) \leq \frac{Z(y)}{a(y)} + \log(a(y)) - 1$, which is tight when $a(y) = Z(y)$. This results in a Tractable Unnormalized version of I_{BA} :

$$\begin{aligned}
I &\geq I_{UBA} \geq \mathbb{E}_{p(x,y)} [f(x, y)] \\
&\quad - \mathbb{E}_{p(y)} \left[\frac{\mathbb{E}_{p(x)} [e^{f(x,y)}]}{a(y)} + \log(a(y)) - 1 \right] \\
&\triangleq I_{TUBA}
\end{aligned}$$

2.2.4. I_{NWJ}

Letting $a(y) = e$ recovers the bound of Nguyen, Wainwright, and Jordan in 2010, also known as f -GAN and [MINE- \$f\$](#) :

$$\mathbb{E}_{p(x,y)}[f(x,y)] - e^{-1}\mathbb{E}_{p(y)}[Z(y)] \triangleq I_{\text{NWJ}}, \quad (5)$$

which no longer needs to learn $a(y)$. But $f(x,y)$ needs to learn to self-normalize. And the optimal critic comes from:

$$\frac{p(x)}{e} e^{f(x,y)} = p(x | y). \quad (6)$$

And the solution is:

$$f^*(x,y) = 1 + \log \frac{p(x | y)}{p(x)}. \quad (7)$$

In conclusion, these unnormalized bounds are unbiased while exhibit a high variance because the log partition function is of high variance.

2.3. Multi-sample unnormalized Lower Bounds

Multi-sample unnormalized bounds have low-variance but high-bias. We will mainly talk about the [InfoNCE](#).

Assume x_1 and y are from a sample pair $p(x_1)p(y | x_1)$ while other $K - 1$ samples are from an independent distribution $x_{2:K} \sim r^{K-1}(x_{2:K})$. We can have:

$$I(X_1; Y) = \mathbb{E}_{r^{K-1}(x_{2:K})} [I(X_1; Y)] = I(X_1, X_{2:K}; Y). \quad (8)$$

Such a multi-sample mutual information can be used in all previous bounds with the same optimal critic $f^*(x_{1:K}, y) = 1 + \log \frac{p(y|x_{1:K})}{p(y)} = 1 + \log \frac{p(y|x_1)}{p(y)}$.

Particularly, if we set the critic in I_{NWJ} to:

$$f(x_{1:K}, y) = 1 + \log \frac{e^{f(x_1,y)}}{a(y; x_{1:K})}, \quad (9)$$

then I_{NWJ} becomes:

$$\begin{aligned}
I(X_1; Y) &\geq I_{\text{NWJ}} \triangleq \mathbb{E}_{p(x,y)}[f(x,y)] - e^{-1} \mathbb{E}_{p(y)}[Z(y)] \\
&= \mathbb{E}_{p(x_{1:K})p(y|x_1)} \left[1 + \log \frac{e^{f(x_1,y)}}{a(y; x_{1:K})} \right] - e^{-1} \mathbb{E}_{p(y)} \left[\mathbb{E}_{p(x_{1:K})} \left[e^{f(x_{1:K},y)} \right] \right] \\
&= 1 + \mathbb{E}_{p(x_{1:K})p(y|x_1)} \left[\log \frac{e^{f(x_1,y)}}{a(y; x_{1:K})} \right] - \mathbb{E}_{p(x_{1:K})p(y)} \left[\frac{e^{f(x_1,y)}}{a(y; x_{1:K})} \right],
\end{aligned}$$

where the inequality comes from the critic not guaranteed to be optimal. The additional samples from $p(x)$ can be used to estimate the partition function $Z(y)$:

$$Z(y) = a(y; x_{1:K}) = m(y; x_{1:K}) = \frac{1}{K} \sum_{i=1}^K e^{f(x_i,y)}. \quad (10)$$

The last term in the bound can be simplified as:

$$\begin{aligned}
\mathbb{E}_{p(x_{1:K})p(y)} \left[\frac{e^{f(x_1,y)}}{m(y; x_{1:K})} \right] &= \frac{1}{K} \sum_{i=1}^K \mathbb{E} \left[\frac{e^{f(x_i,y)}}{m(y; x_{1:K})} \right] \\
&= \mathbb{E}_{p(x_{1:K})p(y)} \left[\frac{\frac{1}{K} \sum_{i=1}^K e^{f(x_i,y)}}{m(y; x_{1:K})} \right] = 1,
\end{aligned} \quad (11)$$

and thus the bound recovers the [InfoNCE](#) loss:

$$I(X; Y) \geq \mathbb{E} \left[\frac{1}{K} \sum_{i=1}^K \log \frac{e^{f(x_i,y_i)}}{\frac{1}{K} \sum_{j=1}^K e^{f(x_i,y_j)}} \right] \triangleq I_{\text{NCE}}. \quad (12)$$

The optimal critic for I_{NCE} is $f(x, y) = \log p(y | x) + c(y)$ like I_{UBA} . Note that I_{NCE} itself is upper bounded by $\log K$, which means that if $I(X; Y) > \log K$, the I_{NCE} bound is loose. The larger the batch, the better the estimation.

2.4. Nonlinearly Interpolated Lower Bounds

We can interpolate between I_{NCE} (high-bias, low-variance) and I_{NWJ} (low-bias, high-variance) to trade off the bias and the variance. Setting the critic to $1 + \log \frac{e^{f(x_1,y)}}{\alpha m(y; x_{1:K}) + (1-\alpha)q(y)}$ with $\alpha \in [0, 1]$ to get a continuum of lower bounds:

$$\begin{aligned}
&1 + \mathbb{E}_{p(x_{1:K})p(y|x_1)} \left[\log \frac{e^{f(x_1,y)}}{\alpha m(y; x_{1:K}) + (1-\alpha)q(y)} \right] \\
&- \mathbb{E}_{p(x_{1:K})p(y)} \left[\frac{e^{f(x_1,y)}}{\alpha m(y; x_{1:K}) + (1-\alpha)q(y)} \right] \triangleq I_\alpha.
\end{aligned} \quad (13)$$

Setting $\alpha = 0$, we can recover I_{NWJ} and $\alpha = 1$, we can recover I_{NCE} . This interpolated bound is upper bounded by $\log \frac{K}{\alpha}$.

2.5. Structured Bounds with Tractable Encoders

In current representation learning area, $p(y | x)$ is usually easily accessible when the y is a learned stochastic representation.

2.5.1. InfoNCE with a Tractable Conditional

An optimal critic of InfoNCE is given by $f(x, y) = \log p(y | x)$. (WARNING: this is strange for me because $c(y)$ is omitted here) We can plug in the optimal f and have a bound:

$$I(X; Y) \geq \mathbb{E} \left[\frac{1}{K} \sum_{i=1}^K \log \frac{p(y_i | x_i)}{\frac{1}{K} \sum_{j=1}^K p(y_i | x_j)} \right]. \quad (14)$$

2.5.2. Leave One Out Upper Bound

In Section 2.1, the upper bound is obtained by a variational distribution $q(y)$.

$$I(X; Y) \leq \mathbb{E}_{p(x)} [KL(p(y | x) \| q(y))], \quad (15)$$

Given a batch of K samples, we can approximate $p(y)$ in this way:

$$p(y) \approx \frac{1}{K} \sum_i p(y | x_i). \quad (16)$$

Leaving out the corresponding y_i for x_i , we can have $q_i(y) = \frac{1}{K-1} \sum_{j \neq i} p(y | x_j)$. Therefore, the upper bound is:

$$I(X; Y) \leq \mathbb{E} \left[\frac{1}{K} \sum_{i=1}^K \left[\log \frac{p(y_i | x_i)}{\frac{1}{K-1} \sum_{j \neq i} p(y_i | x_j)} \right] \right]. \quad (17)$$

With Eq. (14) and (17), we successfully sandwich the MI without introducing a variational distribution. The only difference between these two bounds is whether $p(y_i | x_i)$ is included in the denominator.

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